SERIES TEMPORELLES UNIVARIEES_TP

Diane ANATO & Patrice KOLIE

```
# Importation des librairies
library(readxl)
```

```
Warning: le package 'readxl' a été compilé avec la version R 4.4.3
```

```
library(dplyr)
```

```

Attachement du package : 'dplyr'

Les objets suivants sont masqués depuis 'package:stats':

    filter, lag

Les objets suivants sont masqués depuis 'package:base':

    intersect, setdiff, setequal, union
```

```
library(forecast)
```

```
Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
```

```
library(lmtest)
```

```
Le chargement a nécessité le package : zoo
```

```
Attachement du package : 'zoo'

Les objets suivants sont masqués depuis 'package:base':

    as.Date, as.Date.numeric
```

```
#Importation de la base
BrentSpot <- read_excel("BrentSpot.xls", sheet = "Data 1",skip =2)
```

```
# Renommer la colonne Prix

BrentSpot <- BrentSpot |>
  rename(Pt = `Europe Brent Spot Price FOB (Dollars per Barrel)`)

# Vérification
head(df)
```

```
1 function (x, df1, df2, ncp, log = FALSE)
2 {
3     if (missing(ncp))
4         .Call(C_df, x, df1, df2, log)
5     else .Call(C_dnf, x, df1, df2, ncp, log)
6 }
```

```
# Création du prix retardé
BrentSpot <- BrentSpot |>
  mutate(Pt_1 = lag(Pt, 1))

# Vérification
head(BrentSpot)
```

```
# A tibble: 6 x 3
   Date    Pt  Pt_1
  <dbl> <dbl> <dbl>
1 31917  18.6  NA
2 31918  18.4  18.6
3 31919  18.6  18.4
4 31922  18.6  18.6
5 31923  18.6  18.6
6 31924  18.6  18.6
```

```r
# Création de la colonne rendement
BrentSpot <- BrentSpot |>
  mutate(rt = log(Pt / Pt_1))

# Vérification
head(BrentSpot)
```

```
# A tibble: 6 x 4
   Date    Pt  Pt_1       rt
  <dbl> <dbl> <dbl>    <dbl>
1 31917  18.6  NA   NA
2 31918  18.4  18.6 -0.00971
3 31919  18.6  18.4  0.00541
4 31922  18.6  18.6  0.00269
5 31923  18.6  18.6  0.00161
6 31924  18.6  18.6 -0.00161
```

Estimation des processus autorégressifs via AR(1) et AR(2) par MCO

- $\text{AR}(1) : r_t = \phi_0 + \phi_1 r_{t-1} + \epsilon_t$

```r
# Création du retard 1
BrentSpot <- BrentSpot |>
  mutate(r_1 = lag(rt, 1)) |>
  filter(!is.na(r_1))

# Régression MCO AR(1)
MCO_ar1 <- lm(rt ~ r_1, data = BrentSpot)
summary(MCO_ar1)
```

```
Call:
lm(formula = rt ~ r_1, data = BrentSpot)

Residuals:
     Min       1Q   Median       3Q      Max
-0.36255 -0.01179  0.00029  0.01221  0.18051

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept) 0.0002770  0.0002911   0.951   0.3414
r_1         0.0304985  0.0124556   2.449   0.0144 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02336 on 6440 degrees of freedom
Multiple R-squared:  0.0009301, Adjusted R-squared:  0.000775
F-statistic: 5.996 on 1 and 6440 DF,  p-value: 0.01437
```

On peut voir que r_{t_1} est significativement différent au seuil de 5%.

- AR(2) : $r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 * r_{t-2} + \epsilon_t$

```r
# Création du retard 2
BrentSpot <- BrentSpot |>
  mutate(r_2 = lag(rt, 2)) |>
  filter(!is.na(r_2))

# Régression MCO AR(2)
MCO_ar2 <- lm(rt ~ r_1 + r_2, data = BrentSpot)
summary(MCO_ar2)
```

```
Call:
lm(formula = rt ~ r_1 + r_2, data = BrentSpot)

Residuals:
     Min       1Q   Median       3Q      Max
-0.36267 -0.01171  0.00033  0.01224  0.18084

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.0002788  0.0002912   0.957   0.3384
r_1          0.0308503  0.0124636   2.475   0.0133 *
r_2         -0.0111617  0.0124642  -0.896   0.3706
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02337 on 6437 degrees of freedom
Multiple R-squared:  0.001055,  Adjusted R-squared:  0.0007449
F-statistic:   3.4 on 2 and 6437 DF,  p-value: 0.03343
```

On peut voir que r_{t_1} est significativement différent au seuil de 5% et $r_{t_2} ne l'et pas.

Estimation eà travers la fonction arima

```
# Estimation de AR(1)
fit_ar1 <- Arima(BrentSpot$rt, order = c(1,0,0), include.mean = TRUE)
summary(fit_ar1)
```

```
Series: BrentSpot$rt
ARIMA(1,0,0) with non-zero mean

Coefficients:
         ar1   mean
      0.0305  3e-04
s.e.  0.0125  3e-04

sigma^2 = 0.000546:  log likelihood = 15054.5
AIC=-30103   AICc=-30103   BIC=-30082.69

Training set error measures:
                      ME       RMSE        MAE MPE MAPE      MASE         ACF1
Training set 7.631942e-08 0.02336318 0.01630536 NaN  Inf 0.7031154 0.0003451395
```

```
#Estimation de AR (2)
fit_ar2 <- Arima(BrentSpot$rt, order = c(2,0,0), include.mean = TRUE)
summary(fit_ar2)
```

```
Series: BrentSpot$rt
ARIMA(2,0,0) with non-zero mean

Coefficients:
         ar1      ar2   mean
      0.0308  -0.0112  3e-04
s.e.  0.0125   0.0125  3e-04

sigma^2 = 0.000546:  log likelihood = 15054.9
AIC=-30101.81   AICc=-30101.8   BIC=-30074.72

Training set error measures:
                      ME       RMSE        MAE MPE MAPE      MASE
Training set 9.339399e-09 0.02336172 0.01630193 NaN  Inf 0.7029678
                     ACF1
Training set -0.0002650638
```

```r
# coefficients de AR(1)
coef(MCO_ar1) #MCO
```

```
 (Intercept)          r_1
0.0002769547 0.0304985032
```

```r
coef(fit_ar1) #ARIMA
```

```
         ar1    intercept
0.0305050865 0.0002843709
```

En considérant l'estimation du AR(1), on constate que les coefficient du retard r_{t-1} et la constance(itercept) sont assez proches. Nous avons, 0,O304 avec les MCO et 0,0305 avec la méthode de vraisemblancepour le premier retard.

```r
# coefficients de AR(2)
coef(MCO_ar2)
```

```
 (Intercept)          r_1          r_2
 0.000278792  0.030850317 -0.011161722
```

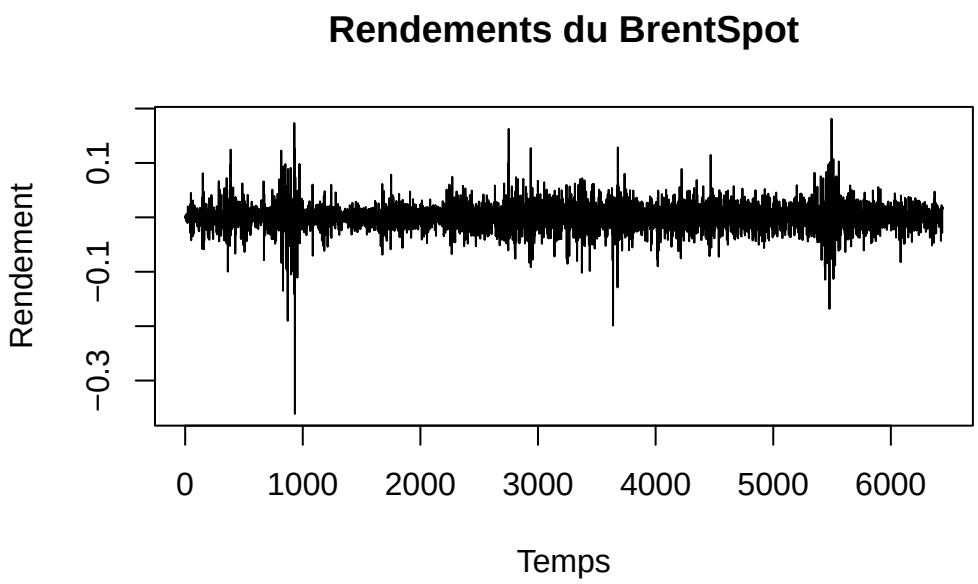
```r
coef(fit_ar2)
```

```
          ar1           ar2     intercept
 0.0308445760 -0.0111588799  0.0002843802
```

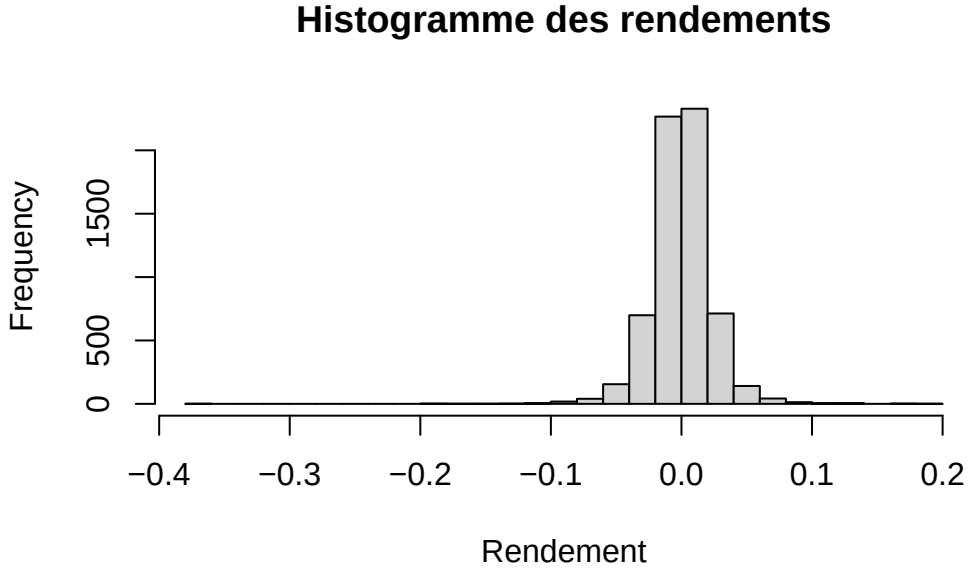
En considérant l'estimation du AR(2), on constate que les coefficient des retards et la constance(itercept) par la méthode des MCO et par la méthode de vraisemblance sont assez proches.

Modélisation ARMA du rendement d'un indice boursier

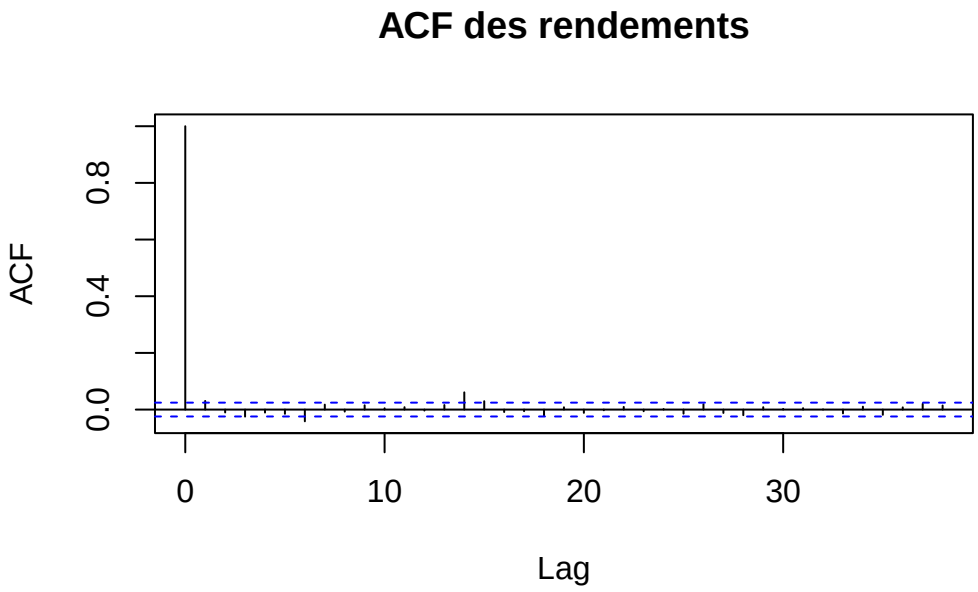
```r
plot(BrentSpot$rt, type="l", main="Rendements du BrentSpot", ylab="Rendement", xlab="Temps")
```



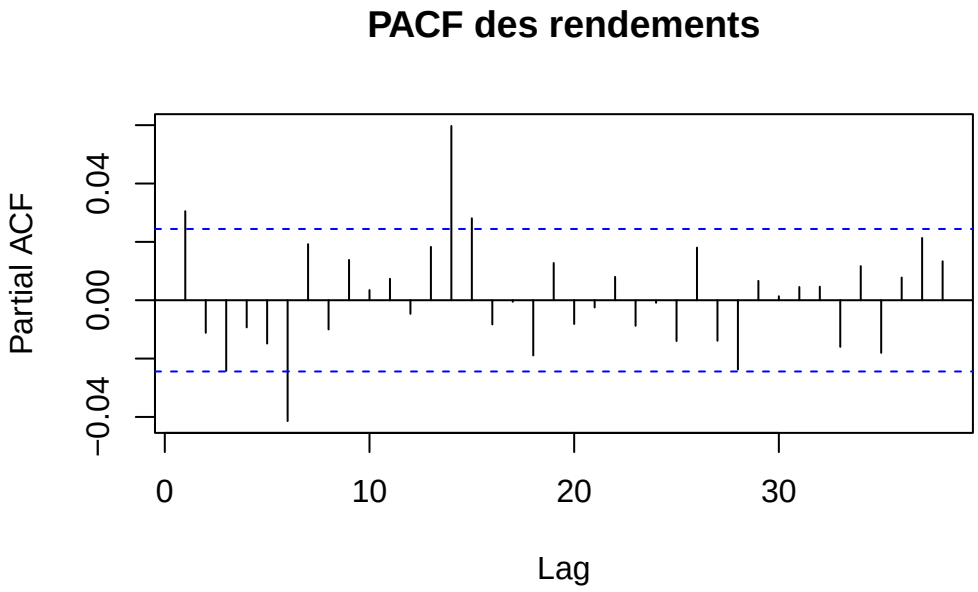

```r
hist(BrentSpot$rt, breaks=30, main="Histogramme des rendements", xlab="Rendement")
```



```r
acf(BrentSpot$rt, main="ACF des rendements")
```




```r
pacf(BrentSpot$rt, main="PACF des rendements")
```



On remarque une décroissance exponentielle au niveau de l'ACF. Et une irrégularité des retards au niveau de la PACF.

```r
library(ggplot2)

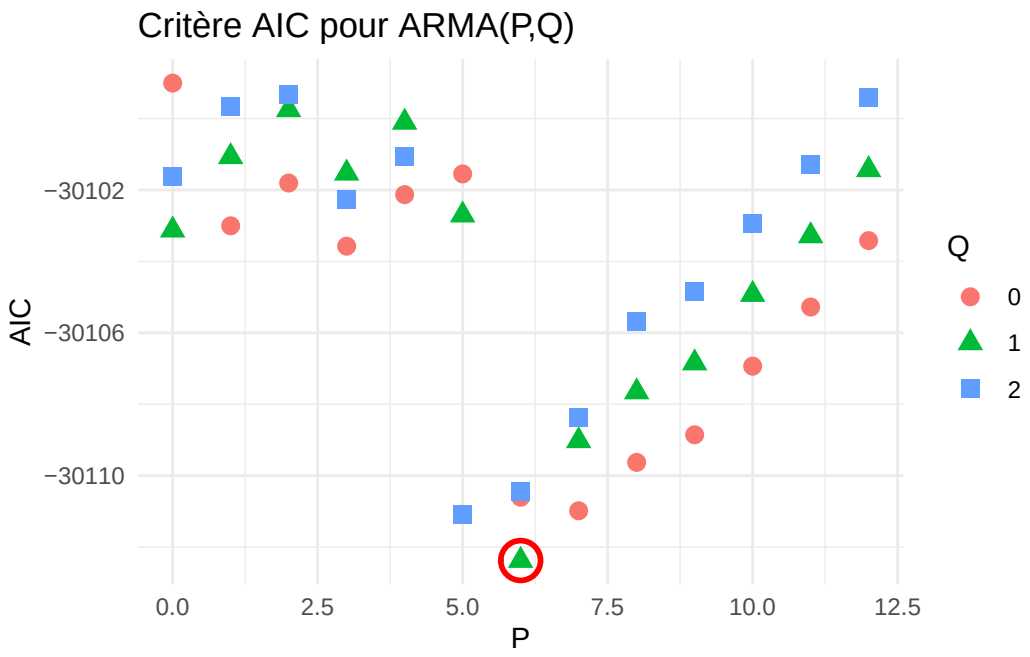
y <- BrentSpot$rt[!is.na(BrentSpot$rt)]

results <- data.frame()

for(p in 0:12){
  for(q in 0:2){
    mod <- arima(y, order = c(p,0,q))
    results <- rbind(results,
                     data.frame(P = p,
                                Q = q,
                                AIC = AIC(mod),
                                BIC = BIC(mod)))
  }
}
```

```r
best <- results[which.min(results$AIC), ]

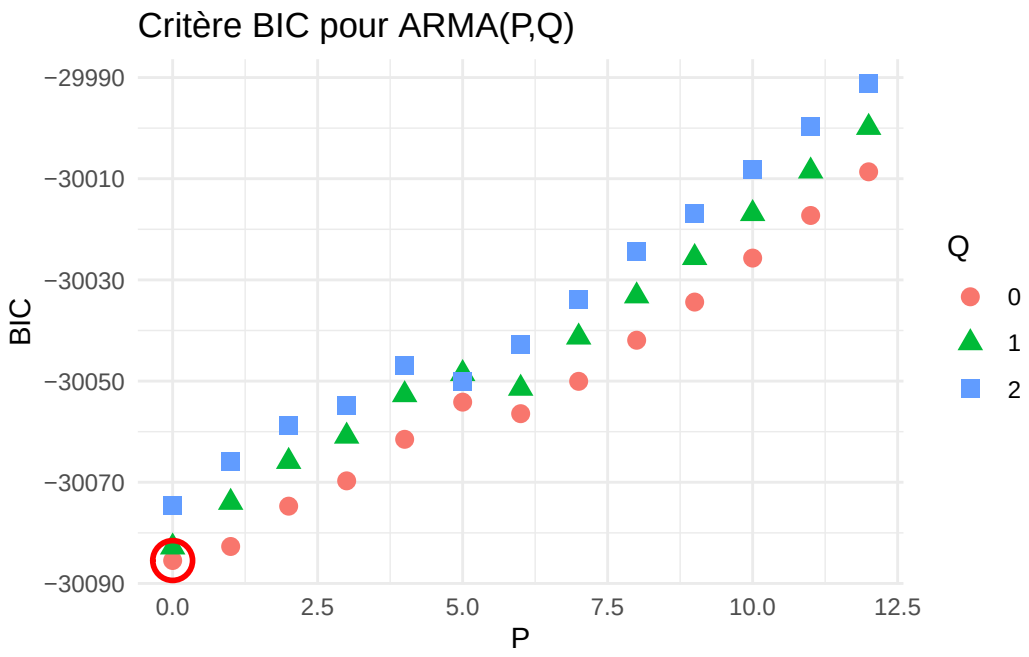
ggplot(results, aes(x = P, y = AIC, shape = factor(Q), color = factor(Q))) +
  geom_point(size = 3) +
  # Cercle rouge autour du meilleur AIC
  geom_point(data = best, aes(x = P, y = AIC),
             shape = 21, fill = NA, color = "red", size = 6, stroke = 1.5) +
  labs(title = "Critère AIC pour ARMA(P,Q)",
       x = "P",
       y = "AIC",
       shape = "Q",
       color = "Q") +
  theme_minimal()
```



Le modèle préconisé est ARMA(6,1)

```
best_bic <- results[which.min(results$BIC), ]

ggplot(results, aes(x = P,
                    y = BIC,
                    shape = factor(Q),
                    color = factor(Q))) +
  geom_point(size = 3) +
  # Cercle rouge autour du meilleur BIC
  geom_point(data = best_bic, aes(x = P, y = BIC),
             shape = 21, fill = NA, color = "red", size = 6, stroke = 1.5) +
  labs(title = "Critère BIC pour ARMA(P,Q)",
       x = "P",
       y = "BIC",
       shape = "Q",
       color = "Q") +
  theme_minimal()
```



Selon le critère BIC, le modèle préconisé est un ARMA(0,2) qui est le MA(2)